

Siemens S700K Point Machine

Switch Blade Adjustment Procedure

Inspection, Measurement and Adjustment of Switch Blade Gap

Document ID:	MAN-PM-001
Asset Type:	Point Machine
Language:	EN
Revision:	Rev C 2025-01-15
Status:	APPROVED FOR USE

1. Scope and Safety

This procedure covers inspection, measurement, and adjustment of the switch blade gap on Siemens S700K and S700H point machines on mainline and yard tracks. Track Protection Procedures (TPP) must be active throughout. Only technicians holding a current Siemens S700K Level-2 Maintenance Certificate may perform this work.

WARNING: Ensure the track section is fully isolated under TPP before beginning any work. Failure to comply may result in serious injury or death from moving train traffic.

1.1 Tools and Materials Required

- Calibrated feeler gauge set, 0.1 mm - 10 mm (calibration certificate required, within 6 months)
- Torque wrench, 20 - 80 Nm range (calibration within 6 months)
- Drive rod adjustment tool P/N S700K-ADJ-02
- 27 mm open-end spanner
- Siemens LF-200 grease cartridge (if lubrication is required during adjustment)
- PPE: high-visibility vest, safety boots, hard hat, gloves
- Camera or mobile device for photographic record

2. Switch Blade Gap Specifications

The switch blade gap is the clearance between the switch blade toe and the stock rail when the point machine has completed a full throw and the locking indicator confirms the locked position. Measure at three points along the blade: 100 mm from the toe, at the mid-point, and 100 mm from the heel. Record all three readings.

Condition	Gap Value	Required Action
Normal - serviceable	0.0 - 3.4 mm	Record and continue in service
Pre-emptive adjustment	3.5 - 3.9 mm	Schedule adjustment within 7 days
Mandatory adjustment	4.0 mm (at limit)	Adjust within 48 hours
Out of service	> 4.0 mm	Withdraw immediately - Priority 1 defect
Drive rod play (acceptable)	< 1.0 mm	No action required
Drive rod play (excessive)	> 1.0 mm	Replace drive rod assembly

WARNING: A confirmed gap exceeding 4 mm creates a risk of wheel flange damage and potential derailment. The asset MUST be immediately withdrawn from service and a Priority 1 work order raised before the track is returned to traffic.

*[Schematic: Switch blade gap measurement positions - toe, mid, heel]
Figure: Switch blade gap measurement positions - toe, mid, heel*

3. Adjustment Procedure

If the measured gap requires adjustment, proceed through all steps in sequence. Do not return the asset to service until Section 4 verification is complete.

- Step 1: Using the 27 mm spanner, break the locking nut on the drive rod turnbuckle. Nominal break-free torque should be approximately 45 Nm. Record if it breaks free at a significantly lower value - this indicates prior under-torque and must be noted.
- Step 2: Turn the turnbuckle clockwise (viewed from motor end) to reduce the gap, or counter-clockwise to increase it. Each full rotation changes blade position by approximately 1.5 mm.
- Step 3: Re-measure the gap at all three points. Variation across measurement points must not exceed +/- 0.3 mm. If variation is greater, inspect the blade for twist or bend before continuing.
- Step 4: When the gap is within 1.0 - 3.4 mm across all measurement points, hold the turnbuckle and re-torque the locking nut to exactly 45 Nm.
- Step 5: Re-measure gap after torquing. If the gap shifted by more than 0.2 mm during locking, the threads are likely worn - replace the turnbuckle assembly and repeat.

NOTE: Apply a thin coat of Siemens LF-200 grease to the drive rod sliding surfaces if the rod shows signs of dry running (bright, unlubricated contact marks). Do not over-grease as excess grease attracts ballast and causes premature wear.

4. Verification and Sign-off

After adjustment, complete all verification steps below before returning to service. Record results on Maintenance Record Sheet MRS-S700K-05 and upload to the CMMS.

- Perform 5 complete switch operations (normal-to-reverse and reverse-to-normal). Switching time must be within 4.0 - 5.5 seconds on all 5 operations.
- Observe motor current draw during switching. Steady-state current must be 2.1 - 2.8 A. Peak inrush current up to 4.5 A for the first 150 ms is normal and acceptable.
- Re-measure blade gap after the 5 operations to confirm no settling has occurred.
- Photograph all three gap measurement points before and after adjustment. Tag with asset ID, timestamp, and technician ID. Attach to the work order in CMMS.

NOTE: If the gap shifts by more than 0.5 mm after 5 verification operations, the turnbuckle or locking nut threads may be worn. Escalate to a Level-3 engineer before returning the asset to service.